



PROJECT

AUTO ANALYTICS PRO



DATA ANALYST



PROJECT INFORMATION

This is a relational database system for a car dealership, built with MySQL. It tracks inventory, sales, customers, maintenance, and manufacturers with 7 interconnected tables. Key features include sales analytics, inventory status monitoring, service history, and customer purchase records. The database supports business operations like identifying popular models, calculating sales commissions, and managing service schedules. Designed for scalability, it can handle complex queries across 50+ sample records. Ideal for automotive CRM systems or dealership management tools.



List all car models along with their manufacturers' names and countries.

```
SELECT
    name, country
FROM
    manufacturers;
```

	name	country
▶	Toyota	Japan
	Honda	Japan
	Ford	USA
	BMW	Germany
	Mercedes-Benz	Germany
	Hyundai	South Korea
	Tesla	USA



Show all customers from Texas (TX) sorted by last name.

```
SELECT
    first_name, last_name, state
FROM
    customers
WHERE
    state = 'TX';
```

	first_name	last_name	state
+	Sarah	Brown	TX
	Robert	Miller	TX



Display all electric cars (fuel_type = 'Electric')
with their prices.

```
SELECT
    car_id, name, fuel_type
FROM
    cars AS c
    JOIN
    manufacturers AS m ON m.manufacturer_id = c.manufacturer_id
WHERE
    fuel_type = 'Electric';
```

	car_id	name	fuel_type
▶	110	Tesla	Electric
	111	Tesla	Electric



Find all sales made in the first quarter of 2023
(January to March).

```
SELECT
    sale_date, first_name, sale_price, payment_method
FROM
    sales AS s
    JOIN
    customers AS c ON c.customer_id = s.customer_id
WHERE
    sale_date BETWEEN '2023-1-1' AND '2023-4-1';
```

	sale_date	first_name	sale_price	payment_method
▶	2023-01-15	John	24500.00	Loan
	2023-02-20	Emily	35500.00	Cash
	2023-03-10	Michael	45000.00	Lease

Calculate the total sales amount for each salesperson.

```
SELECT
    first_name, SUM(sale_price) AS sale
FROM
    sales AS s
    JOIN
    customers AS c ON s.customer_id = c.customer_id
GROUP BY first_name;
```

	first_name	sale
	John	53000.00
	Emily	35500.00
	Michael	45000.00
	Sarah	59000.00
	David	22500.00
	Jessica	54000.00
	Robert	89000.00



Find the average price of cars by manufacturer, ordered from highest to lowest.

```
SELECT
    AVG(sale_price) AS average_price
FROM
    sales;
```

	average_price
▶	44750.000000



Show all cars that are currently available in inventory (current_status = 'Available').

```
SELECT
    name, current_status
FROM
    inventory AS i
    JOIN
    cars AS c ON i.car_id = c.car_id
    JOIN
    manufacturers AS m ON m.manufacturer_id = c.manufacturer_id
WHERE
    current_status = 'Available';
```

	name	current_status
▶	Honda	Available
	Ford	Available
	Hyundai	Available
	BMW	Available



LIST CUSTOMERS WHO PURCHASED MORE THAN ONE CAR (APPEAR MORE THAN ONCE IN SALES TABLE).

```
SELECT
    s.customer_id,
    cu.first_name,
    SUM(s.sale_price) AS total_spent,
    COUNT(s.customer_id) AS purchase_count
FROM sales AS s
JOIN cars AS c ON s.car_id = c.car_id
JOIN customers AS cu ON cu.customer_id = s.customer_id
JOIN manufacturers AS m ON m.manufacturer_id = c.manufacturer_id
GROUP BY s.customer_id, cu.first_name;
```

	customer_id	first_name	total_spent	purchase_count
▶	1001	John	53000.00	2
	1002	Emily	35500.00	1
	1003	Michael	45000.00	1
	1004	Sarah	59000.00	1
	1005	David	22500.00	1
	1006	Jessica	54000.00	1
	1007	Robert	89000.00	1



FIND THE TOP 3 MOST EXPENSIVE CARS SOLD, INCLUDING CUSTOMER INFORMATION.

```
SELECT
    s.customer_id,
    cu.first_name,
    cu.last_name,
    name,
    sale_price,
    cu.email,
    cu.phone,
    cu.address,
    cu.city,
    cu.state
FROM
    sales AS s
    JOIN
    cars AS c ON s.car_id = c.car_id
    JOIN
    customers AS cu ON cu.customer_id = s.customer_id
    JOIN
    manufacturers AS m ON m.manufacturer_id = c.manufacturer_id
ORDER BY sale_price DESC
LIMIT 3;
```

	customer_id	first_name	last_name	name	sale_price	email	phone	address	city	state
▶	1007	Robert	Miller	Tesla	89000.00	robert.m@email.com	555-789-0123	159 Birch Blvd	San Antonio	TX
	1004	Sarah	Brown	BMW	59000.00	sarah.b@email.com	555-456-7890	321 Elm St	Houston	TX
	1006	Jessica	Garcia	Mercedes-Benz	54000.00	jessica.g@email.com	555-678-9012	987 Cedar Ln	Philadelphia	PA



find avrage car price

```
SELECT  
    ROUND(AVG(price), 2) as avg_price  
FROM  
    cars;
```

	avg_price
▶	42166.66



Find salespersons who have sold cars with prices above the average car price.

```
SELECT
    name, s.sale_price
FROM
    salespersons AS sp
    JOIN
    sales AS s ON sp.salesperson_id = s.salesperson_id
WHERE
    s.sale_price > 42166.66;
```

	name	sale_price
▶	Alex Turner	45000.00
	Morgan Lee	59000.00
	Alex Turner	54000.00
	Morgan Lee	89000.00



Show customers who bought cars from manufacturers based in their same country.

```
SELECT
  c.first_name, city
FROM
  sales AS s
  JOIN
  customers AS c ON s.customer_id = c.customer_id
  JOIN
  cars ON cars.car_id = s.car_id
  JOIN
  manufacturers AS m ON m.manufacturer_id = cars.manufacturer_id
WHERE
  cars.manufacturer_id = m.manufacturer_id;
```

first_name	city
John	New York
John	New York
Emily	Los Angeles
Michael	Chicago
Sarah	Houston
David	Phoenix
Jessica	Philadelphia
Robert	San Antonio



Calculate the total commission earned by each salesperson.

```
SELECT
    sp.salesperson_id,
    sp.name AS salesperson_name,
    sp.commission_rate,
    (SELECT COUNT(*) FROM sales s WHERE s.salesperson_id = sp.salesperson_id) AS total_sales,
    (SELECT COALESCE(SUM(sale_price), 0) FROM sales s WHERE s.salesperson_id = sp.salesperson_id) AS total_sales_amount,
    ROUND((SELECT COALESCE(SUM(sale_price * sp.commission_rate), 0)
           FROM sales s
           WHERE s.salesperson_id = sp.salesperson_id), 2) AS total_commission
FROM
    salespersons sp
ORDER BY
    total_commission DESC;
```

	salesperson_id	salesperson_name	commission_rate	total_sales	total_sales_amount	total_commission
▶	4	Morgan Lee	0.15	2	148000.00	22200.00
	1	Alex Turner	0.10	4	152000.00	15200.00
	2	Jamie Wilson	0.12	1	35500.00	4260.00
	3	Taylor Smith	0.08	1	22500.00	1800.00



Find the most popular car color among sold vehicles.

```
SELECT
    color, COUNT(color) AS puplor_color
FROM
    cars AS c
    JOIN
    sales AS s ON c.car_id = s.car_id
GROUP BY color
ORDER BY puplor_color DESC
LIMIT 2;
```

	color	puplor_color
	Silver	2
	Blue	2



THANK YOU

